• Representation of real numbers on ~~[illegible]~~ number line/
Geometrical representation.

Repeating questions

$\sqrt{5}$ =

= 2.234	2.236
2	$\overline{5}.\overline{00}\,\overline{00}\,\overline{00}$
+2	4
42	1 00
+2	84
443	16 00
+3	1329
4466	27100
+6	26796
	....

$\sqrt{24}$ = 4.898

= 4.90 4	$\overline{24}.\overline{00}\,\overline{00}\,\overline{00}$
= 4.9 +4	16
88	800
+8	704
969	9600
+9	8721
9788	879 00
+8	78304
	.....

$\sqrt{13}$ = 3.605

	3.605
3	$\overline{13}.\overline{00}\,\overline{00}\,\overline{00}$
+3	9
66	4 00
+6	396
7205	4 0000
+5	36025
	....

3	$\overline{13}.\overline{00}\,\overline{00}\,\overline{00}$
+3	9
66	4 00
+6	396
7205	4 00 00
	36025

Homework

Q1. (3 rational no. between $\sqrt{15}$ and $\sqrt{16}$.)

Ans =

$\sqrt{15}$ = 3.872 ...

$\sqrt{16}$ = 4

• 1 → 3.873
• 2 → 3.874
• 3 → 3.875

Q2. (3 irrational no. between $\sqrt{2}$ and $\sqrt{5}$)

Ans =

$\sqrt{2}$ = 1.414

$\sqrt{5}$ = 2.236

• 1 → 1.415010010001....
• 2 → 1.416010010001....
• 3 → 1.417010010001....

Q3. (4 rational no. between $\sqrt{2}$ and 3.

Ans →

$\sqrt{2}$ = 1.414

and 3

• 1.414 — 3
• 1 → 1.4512
• 2 → 1.513
• 3 → 1.514
• 4 → 1.515

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Q4.

Ans:- 1. $\sqrt{13}$ = 3.605

	3.605
3	$\overline{13}\cdot\overline{00}\,\overline{00}\,\overline{00}$
+3	9
66	400
+6	396
7205	40000
+5	36005
7210	3995...
	.....

2. $\sqrt{23}$

4.795

	4.795
4	$\overline{23}\cdot\overline{00}\,\overline{00}\,\overline{00}$
+4	16
87	700
+7	609
949	9100
+9	8541
9585	55900
+5	47925
	7975

3. $\sqrt{24}$

4.9

	4.898
4	$\overline{24}\cdot\overline{00}\,\overline{00}\,\overline{00}$
+4	16
88	800
+8	704
969	9600
+9	8721
9788	87900
+8	78304

$\div 4.90 \rightarrow 4.9$

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$\sqrt{52} = 7.211$

	7.211
7	$\overline{52}\,\overline{00}\,\overline{00}\,\overline{00}$
+7	49
142	300
+2	284
1441	1600
+1	1441
14421	15900
+1	14421
14422	1479
	.....

	7.211
7	$\overline{52}\cdot\overline{00}\,\overline{00}\,\overline{00}$
+7	49
142	300
+2	284
1441	1600
+1	1461

$\overline{52}\cdot\overline{00}\,\overline{00}\,\overline{00}$

142 2 — 1467 2

284

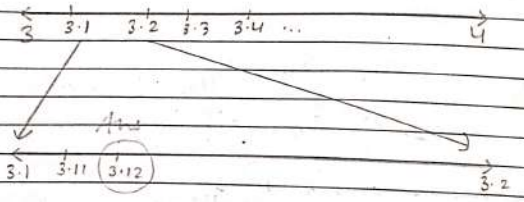
143 3 1

429

Spiral

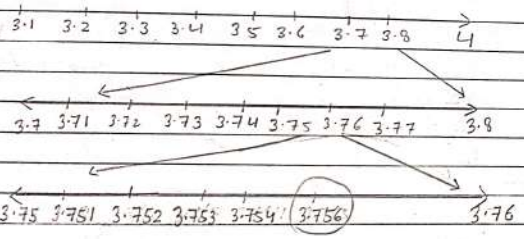
18 May case-I → terminating decimal expansion.

• Represent 3.12 on number line.



Ex-1.4

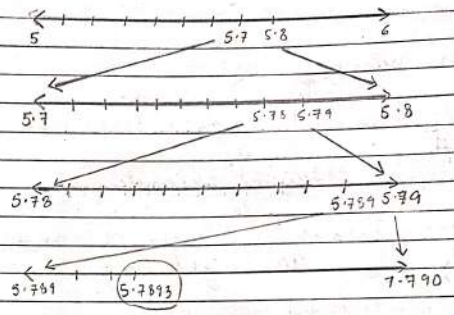
Q1.) 3·756 on number line.



H.W

Eg-11

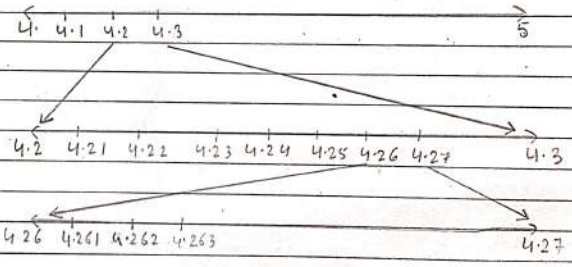
Q. Visualise 5·7893 on number line.



Ex-1.4

Q2. Visualise $4.\overline{26}$ on numberline (upto 4 decimal places)

∴ 4.2626



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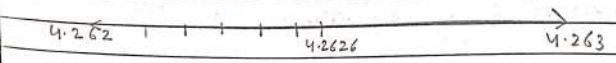


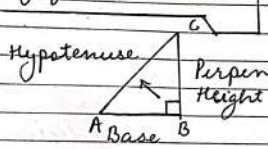
Case-III

* Represent $\sqrt{2}$ on number line: –

↓ Hypotenuse

* 1 = Unit
↓
cm, m
↓
any fixed distance

↓ use → Pythagoras theorem
↓
In right triangle.
↓
Triangle in which one angle 90° = Right angle



* Pythagoras theorem

$\Rightarrow H^2 = B^2 + P^2$

$\Rightarrow$ sq. root

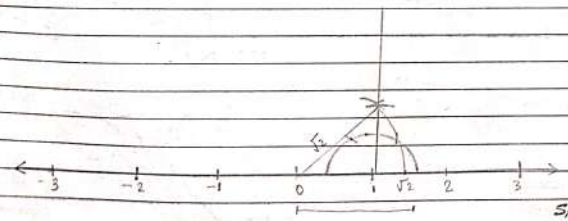
$H = \sqrt{B^2 + P^2}$

$\sqrt{2} = \sqrt{1^2 + 1^2}$

$\sqrt{2} = \sqrt{1+1}$

$\sqrt{2} = \sqrt{2}$

• $\sqrt{2}$.

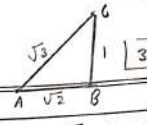


Spiral

$(\sqrt{3})^2 = (\sqrt{2})^2 + (1)^2$

$3 = 2 + 1$

$\boxed{3 = 3}$



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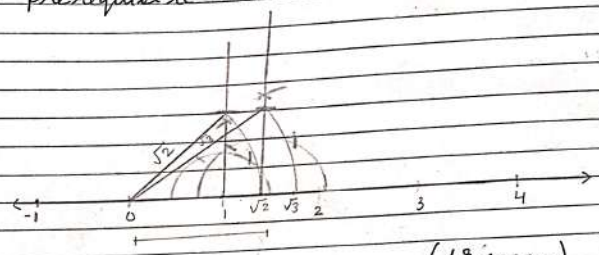
$\sqrt{3}$. → prerequisite → $\sqrt{2}$ needed



(18 may)

Homework

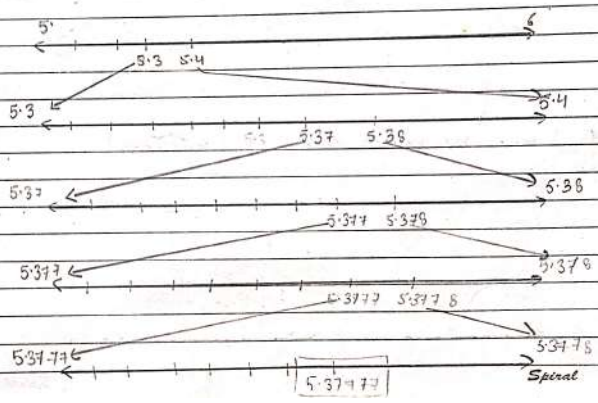
~~Ex-1.4~~

~~Q7~~

Eg-11

Q. ~~4~~ $5.3\overline{7}$ (visualise on number line) [upto 5 decimal places]

Ans. - 5.37777



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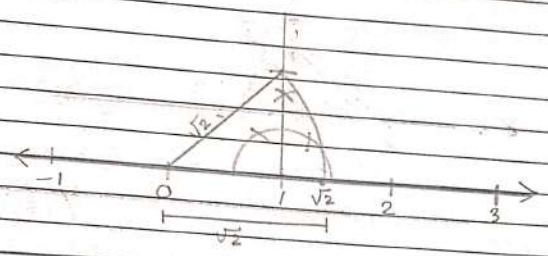
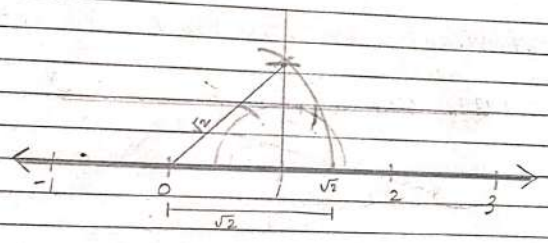
Repeated question-

1. $\sqrt{2}$ on no. line (2 times)

• 1st time

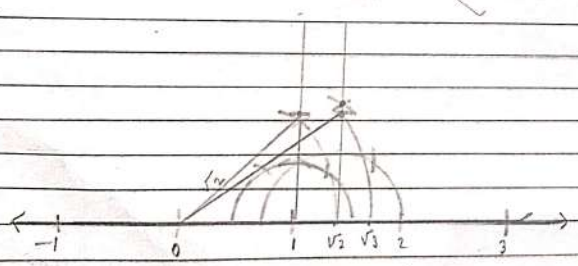


• 2nd time

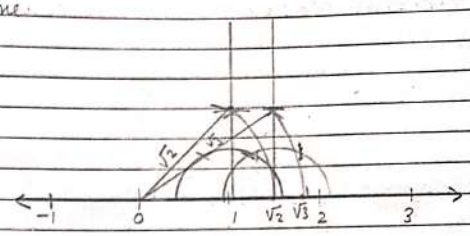


2. root 3 on number line (2 times)

1st time.

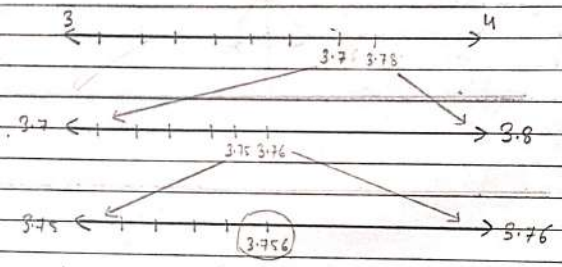


2nd time.



Q. 3.756 on number line (visualise)

Ans-



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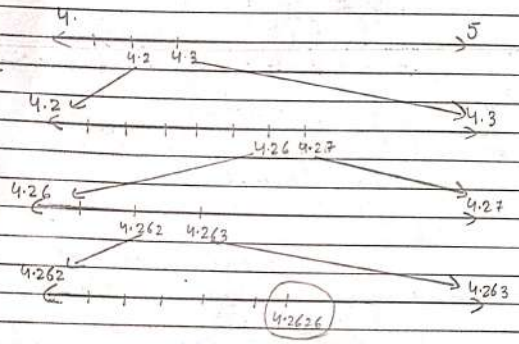
Q. Visualise $4.\overline{26}$ on number line ~~(Repeated quest)~~
(upto 4 decimal places)

Ans- ∵ 4.2666



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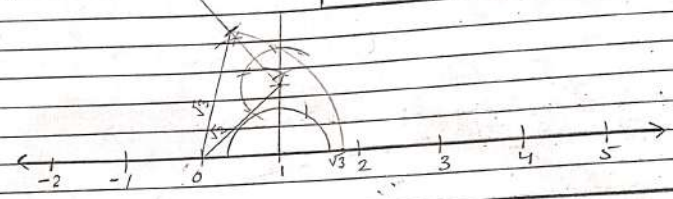
$(H)^2 = B^2 + P^2$
$(\sqrt{5})^2 = \sqrt{2^2 + 1^2}$
$\sqrt{5} = \sqrt{5}$

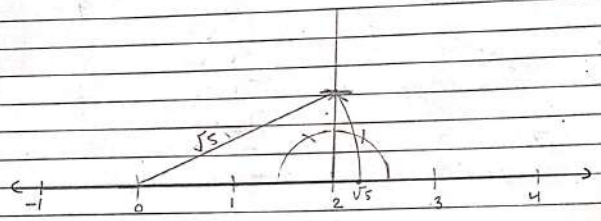
19 May, 22

2nd method

$\sqrt{3}$



Represent $\sqrt{5}$ on number line.



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